

Technology Stack

A pragmatic Java microservice backend, a responsive Next.js frontend, and isolated AI orchestration.

Backend and Data



Java 21

Primary language for the backend services and shared engineering conventions.



Spring Boot 3.x

Independent REST services with configuration profiles, validation, and application lifecycle support.



Spring Security

JWT-protected API boundaries and stateless security configuration.



PostgreSQL

Relational persistence, managed through JPA/Hibernate and Flyway migrations.



Flyway

Versioned database migration support for service-owned schemas.



Redis

Available through Docker Compose as local platform infrastructure.



Kafka

Available through Docker Compose as local platform infrastructure.



Docker Compose

Starts PostgreSQL, Redis, and Kafka consistently for local development.



Maven

Monorepo build, dependency management, and service-specific execution.

Frontend and Platform Tooling

NX

Next.js

App Router frontend with a same-origin API proxy and protected-route experience.

R

React

Component-based interface for catalog, cart, AI chat, payment, and tracking flows.

TS

TypeScript

Safer frontend contracts and component implementation.

TW

Tailwind CSS

Responsive, utility-led presentation system for the frontend.

OA

OpenAI API

The Responses API is accessed only inside the dedicated AI service.

G5

GPT-5.6

Conversation reasoning, intent detection, follow-up questions, and multilingual translation when configured.

CX

Codex

Accelerated implementation, API integration, security diagnosis, tests, UI iteration, and documentation.

REST

REST APIs

Clear service contracts used by the web client and reusable by future native clients.

JWT

JWT

Auth-issued Bearer tokens authenticated across protected backend services.

API

Swagger / OpenAPI

Interactive API documentation where Springdoc is configured: Auth, Customer, Catalog, Order, and AI.

Why this shape: product facts remain in Catalog, cart operations remain in Cart, checkout and tracking remain in Order, mock payment remains in Payment, and OpenAI interaction remains in AI. The frontend composes those existing APIs instead of duplicating backend logic.

Security and Testing Baseline

JWT + Spring Security

Security is stateless. The frontend proxy forwards the Authorization header, and protected services validate the token issued by Auth.

Quality Tooling

The project uses JUnit 5, Mockito, REST Assured where applicable, frontend lint/build checks, and integration-flow verification across services.